



Evaluating Parameter Sharing for Spoofing-Aware Speaker Verification: A Case Study on the ASVspoof 5 Dataset

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Abstract

Spoofing-aware speaker verification (SASV) is an important but challenging task and has been a primary focus of the recently organized ASVspoof 5 challenge. As SASV integrates automatic speaker verification (ASV) and countermeasure (CM) systems, its performance depends on the effectiveness of each system. This study systematically examines the impact of different parameter-sharing (PS) strategies, which facilitate joint optimization, on SASV performance using the ASVspoof 5 dataset. Experimental results indicate that PS enhances performance for specific attack types and codec conditions. For example, the baseline system achieves a min a-DCF of 0.329 on the A26 attack, which improves to 0.233 with PS. Similarly, for AMR-compressed signals, PS yields a 14.09% performance gain. These observations show that PS techniques are effective in mitigating certain spoofing attacks and improving robustness to degraded audio conditions in SASV systems.

Index Terms: spoofing-aware speaker verification, parameter sharing, ASVspoof

1. Introduction

Automatic Speaker Verification (ASV) systems authenticate a speaker's identity based on their voice [1]. These systems are widely used in biometric authentication [1], voice-controlled applications [2], and security-sensitive domains [3]. Typically, ASV systems extract speaker embeddings from test utterances and compare them with stored reference embeddings [4]. However, they remain vulnerable to spoofing attacks, where adversaries attempt to deceive the system using synthetic, converted, or replayed speech, necessitating countermeasures (CMs) to enhance robustness.

To address these vulnerabilities, the ASVspoof 5 challenges¹ were introduced to advance the development of CM systems. The first four editions (2015 [5], 2017 [6], 2019 [7] and 2021 [8]) primarily focused on stand-alone CM systems for detecting synthetic or replayed speech. Unlike previous editions, ASVspoof 5 (2024) [9] included non-studio quality data, diverse recording conditions, and state-of-the-art text-to-speech (TTS), voice conversion (VC), and adversarial attacks. The challenge featured two tracks: Track 1 focused on speech deepfake detection, while Track 2 addressed spoofing-aware speaker verification (SASV). This study focuses on Track 2 (SASV).

The SASV task aims to verify a speaker's identity while simultaneously detecting spoofing attempts, making it more suitable for real-world applications. Traditional SASV approaches train ASV and CM systems separately, combining their scores at the decision stage. While this ensures both systems contribute

to the final decision, it limits information exchange between ASV and CM, reducing overall effectiveness.

The vast majority of the studies focusing on the SASV task (Track 2 in the ASVspoof 5 challenge) adopt this score-fusion approach, where several independently trained CM and ASV systems generate separate scores for each task (CM and ASV tasks) and then these scores are combined in various ways [10, 11, 12, 13, 14]. In contrast, in [15] a parallel embedding fusion approach was proposed, where ASV and CM embeddings are directly proposed by separate branches in a joint model. Although this method integrates ASV and CM information at the embedding level, the two branches remain independent, preventing mutual knowledge exchange between ASV and CM during training. As a result, ASV does not leverage spoof detection cues from CM, and CM does not benefit from speaker identity-related features learned by ASV, limiting SASV performance.

To overcome this limitation, we propose a parameter-sharing-based approach that facilitates knowledge exchange between ASV and CM while preserving their distinct learning objectives. Unlike independent ASV-CM models, our method introduces parameter-sharing mechanisms at different levels, enabling cross-task feature transfer. This approach is expected to enhance learning efficiency and improve SASV performance by leveraging shared representations.

2. Proposed Method

2.1. Baseline - Parallel Embedding Fusion

In the basic embedding fusion approach, ASV ($e_{ASV}^{enr}, e_{ASV}^{tst}$) and CM (e_{CM}^{tst}) embeddings are concatenated into a single feature vector and processed by a fully connected (FC) DNN classifier to determine the SASV decision $y_{sasv} \in \{0, 1\}$. While this method, similar to the Baseline2 system [16] in SASV2022 challenge, integrates information from both ASV and CM models, a single DNN may struggle to optimally learn SASV-discriminative features due to architectural constraints. Furthermore, preliminary experiments indicate that different combinations of ASV and CM embeddings considerably impact SASV performance, highlighting the need for a more flexible fusion strategy.

To address these limitations, [15] proposes a parallel embedding fusion model, wherein two identical DNNs process ASV and CM embeddings separately but synchronously. Similarly in this work, we adopt the parallel SASV model (Fig. 1(a)) as our baseline system. In this model, ASV and CM embedding pairs ($e_{asv}^{enr}, e_{asv}^{tst}$) and ($e_{cm}^{enr}, e_{cm}^{tst}$) are independently processed by two parallel DNNs, each learning SASV-relevant features from its respective inputs. The networks produce independent SASV probability scores, s_{sasv}^1 and s_{sasv}^2 , which are

¹<https://www.asvspoof.org/>

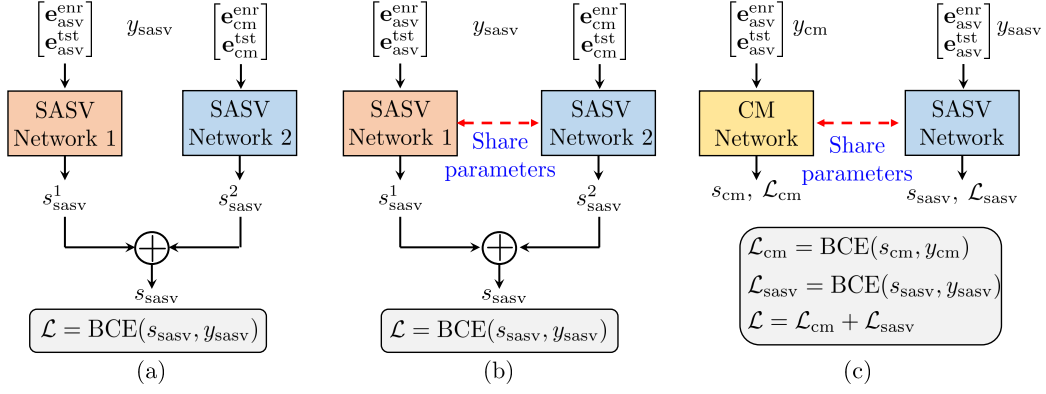


Figure 1: Overview of the proposed spoofing-aware speaker verification (SASV) models. (a) The baseline parallel embedding fusion model. (b) The soft parameter sharing model. (c) The integrated SASV-CM framework. In all these models the networks are identical with four FC layers consisting of 256, 128, 64 and 1 neurons, respectively.

subsequently averaged to obtain the final SASV decision score.

Both parallel networks share the same architecture, consisting of four fully connected layers with 256, 128, 64 and 1 neurons, respectively. The models are jointly trained using the binary cross entropy (BCE) loss, computed based on the SASV decision score s_{sasv} and the corresponding SASV labels y_{sasv} . Despite their identical structure, the distinct input embeddings allow each network to capture complementary information, thereby enhancing the overall robustness of the system while maintaining architectural consistency.

2.2. Soft Parameter Sharing in Parallel SASV Networks

In the soft parameter sharing approach, similar to the baseline system described above, we employ a parallel SASV network in which one branch processes ASV embeddings, and the other processes CM embeddings. The goal is to enhance SASV learning by enabling controlled information exchange between ASV and CM embeddings, allowing both tasks to contribute meaningfully to the final decision. This strategy facilitates knowledge transfer while preserving task-specific representations within a multi-task training framework.

As illustrated in Fig. 1 (b), soft parameter sharing is implemented by introducing trainable connections between the branches that process ASV and CM embeddings separately, hence enabling information flow between them. This design ensures that embeddings from one branch influence the learning process of the other, thereby enriching feature representations for SASV. To assess the effectiveness of this approach, we investigate the following parameter-sharing configurations:

• Setup 1: ASV-to-CM Information Flow

In this setup, knowledge is transferred from the branch processing ASV embeddings to the branch processing CM embeddings, enabling the latter to learn SASV-specific features. Meanwhile, the features learned by the ASV branch remain unchanged. This approach ensures that CM branch integrates SASV features extracted from ASV embeddings, thereby enhancing SASV performance. The parameter sharing mechanism in this setup is formulated as follows:

$$e_{cm}^i := \alpha_1 \times e_{cm}^i + \alpha_2 \times e_{asv}^i, \quad (1)$$

$$e_{asv}^i := e_{asv}^i. \quad (2)$$

Here, e_{asv}^i and e_{cm}^i represent the SASV feature vectors learned by the ASV and CM branches, respectively, at the i th layer ($i = 1, 2, 3, 4$). The learnable parameters α_1 and α_2 control the extent of the parameter sharing, ensuring that the CM branch incorporates relevant information from ASV branch while retaining its own learned features.

• Setup 2: Bidirectional Parameter Sharing

In this setup, information flows in both directions: ASV-to-CM and CM-to-ASV. This mutual exchange enhances the feature representation of both embedding spaces, fostering a more comprehensive SASV model. The bidirectional parameter-sharing mechanism is formulated as follows:

$$e_{asv}^i := \alpha_{11} \times e_{cm}^i + \alpha_{12} \times e_{asv}^i, \quad (3)$$

$$e_{cm}^i := \alpha_{21} \times e_{cm}^i + \alpha_{22} \times e_{asv}^i. \quad (4)$$

The learnable parameters, α_{11} , α_{12} , α_{21} , and α_{22} control the degree of parameter sharing, ensuring a balanced contribution from both embedding spaces.

The final SASV decision score is computed as a weighted sum of the outputs from both branches:

$$s_{sasv} = a_1 s_{sasv}^1 + a_2 s_{sasv}^2, \quad (5)$$

where a_1 and a_2 are trainable parameters, enabling an adaptive weighting mechanism that optimally integrates the information from both branches.

• Setup 3: ASV-Enriched CM Embeddings

In this configuration, the CM branch is enriched by incorporating the SASV features extracted from the ASV branch. This enables the CM branch to leverage complementary information from ASV embeddings while preserving its primary role in CM. By explicitly integrating ASV-derived features, the CM branch gains enhanced robustness against spoofing attacks. The parameter sharing operation for this setup is defined as:

$$e_{cm}^i := \text{concat}(e_{asv}^i, e_{cm}^i), \quad (6)$$

where $i = 1, 2, 3$ represents the FC layer at which parameter sharing is applied. By concatenating CM-derived features with ASV-derived embeddings, this setup facilitates a more comprehensive feature representation, improving the system's ability to distinguish genuine and spoofed audio

Table 1: Comparison of different parameter sharing setups in terms of minimum a -DCF on development (Dev) and evaluation (Eval) sets. The table includes results for soft parameter sharing (Setups 1-4), multi-task learning parameter sharing (Setups 5-7), and score fusion. The score fusion has been applied by averaging of the scores of corresponding rows and columns. The best result is **bolded**.

Layer	Setup 1		Setup 2		Setup 3		Setup 4		Setup 5		Setup 6		Setup 7		Score Fusion	
	Dev	Eval	Dev	Eval	Dev	Eval	Dev	Eval	Dev	Eval	Dev	Eval	Dev	Eval	Dev	Eval
$i = 1$	0.2027	0.6454	0.2894	0.6454	0.2012	0.5576	0.2216	0.5875	0.1978	0.5564	0.1985	0.5565	0.2029	0.5912	0.1834	0.5479
$i = 2$	0.1932	0.6455	0.2753	0.6455	0.1883	0.5534	0.2084	0.5549	0.2024	0.5642	0.2043	0.5731	0.2005	0.5730	0.1756	0.5358
$i = 3$	0.1947	0.6253	0.2667	0.6253	0.1973	0.5584	0.1945	0.5364	0.2044	0.5815	0.2001	0.5680	0.1995	0.5646	0.1773	0.5334
$i = 4$	0.1986	0.5549	0.2705	0.6347	—	—	—	—	—	—	—	—	—	—	0.2031	0.5761
$i = 1, 2, 3$	—	—	—	—	0.1898	0.5596	0.2194	0.5793	0.2064	0.5578	0.2080	0.5473	0.2024	0.5649	0.1779	0.5262
$i = 1, 2, 3, 4$	0.1904	0.5430	—	—	—	—	—	—	—	—	—	—	—	—	0.1904	0.5430
Score Fusion	0.1818	0.5364	0.2584	0.6178	0.1793	0.5343	0.1931	0.5436	0.1798	0.5302	0.1839	0.5290	0.1801	0.5459	—	—

samples while maintaining the speaker-discriminating capability.

- **Setup 4: Statistical CM Integration into ASV Embeddings**

Rather than incorporating raw CM embeddings, this setup extracts statistical features (mean and standard deviation) from the SASV features learned by the CM branch and concatenates them with the features learned from the ASV branch. This method provides auxiliary information while minimizing redundancy:

$$e_{ASV}^i := \text{concat}(e_{ASV}^i, \mu_{CM}^i, \sigma_{CM}^i), \quad (7)$$

where μ_{CM}^i and σ_{CM}^i represent the mean and the standard deviation of the SASV embeddings learned by the CM branch at the i th layer. This configuration helps retain only the most relevant spoof detection characteristics while ensuring that the ASV embeddings remain the primary decision-making component.

2.3. Multi-Task Learning

To enhance the performance of the SASV task, we adopt a multi-task learning framework with parameter sharing. This approach leverages two parallel branches that process ASV embeddings. While one branch tries to learn the spoofing detection task from the ASV embeddings, the other aims at learning the SASV task. Hence, the proposed framework incorporates CM-related features learned from the ASV embeddings. As in the baseline model and parameter sharing approaches described above, each branch consists of three fully connected layers with 256, 128, and 64 nodes, respectively, maintaining the same architectural consistency with baseline parallel fusion model. As illustrated in Fig. 1(c), this design facilitates knowledge transfer between branches, enabling the SASV model to exploit spoof-related features while preserving its speaker verification capabilities.

Both branches receive ASV embeddings as input, however, they differ in the way CM embeddings are integrated. As depicted in Fig. 1(c), the framework employs a multi-task loss function comprising the CM loss ($\mathcal{L}_{CM}$), the SASV loss ($\mathcal{L}_{SASV}$), and the overall objective function ($\mathcal{L} = \mathcal{L}_{CM} + \mathcal{L}_{SASV}$), which balances the contributions of both tasks.

To explore the impact of different fusion strategies, we investigate three integration methods for incorporating CM embeddings learned from the ASV embeddings into the SASV branch:

- **Setup 5: Element-wise Addition of CM Embeddings**

In this setup, CM embeddings are added element-wise to ASV embeddings, allowing the network to refine the ASV

representations with spoof detection information:

$$e_{ASV}^i := e_{SASV}^i + e_{CM}^i \quad (8)$$

where e_{ASV}^i is the SASV features learned from the ASV embeddings at the i th layer and e_{CM}^i is the CM features learned from the ASV embeddings by the CM branch at the i th layer.

- **Setup 6: Concatenation of CM and ASV Embeddings**

In this setup, CM embeddings learned from the ASV features by the CM branch are concatenated with ASV embeddings, enriching the feature space and preserving task-specific information:

$$e_{ASV}^i := \text{concat}(e_{ASV}^i, e_{CM}^i) \quad (9)$$

This concatenation increases the feature dimension at the output of the i th layer, while enabling the network to exploit complementary information from both tasks and enhancing the representation power of the SASV system.

- **Setup 7: Statistical Integration of CM Embeddings**

Instead of directly incorporating raw CM embeddings, we extract statistical descriptors—mean (μ_{CM}) and standard deviation (σ_{CM})—and concatenate them with ASV embeddings:

$$e_{ASV}^i := \text{concat}(e_{ASV}^i, \mu_{CM}, \sigma_{CM}) \quad (10)$$

This method captures essential statistical characteristics of CM embeddings, reducing redundancy while retaining crucial spoof detection cues. This integration ensures a compact yet informative feature representation, improving both generalization and robustness.

3. Experimental Setup

3.1. Dataset

The SASV experiments were conducted using ASVspoof 5 challenge Track 2 dataset [9] under the closed-condition setup, corresponding no external data was used utilized for system development [17]. ASVspoof 5 dataset comprises three disjoint subsets: *training*, *development* and *evaluation*. The training subset was used to train the proposed SASV system, while the development set was employed for hyperparameter tuning. The evaluation subset was reserved for assessing final system performance. Additional details regarding the dataset can be found in [9, 17].

3.2. Model Training and Performance Criterion

In our experiments, we extracted ASV and CM embeddings from both enrollment and test utterances to incorporate speaker

Table 2: Relative improvement (%) in a-DCF for different attack and codec categories using the best-performing configuration of **Setup 4** at $i = 3$. Improvements are computed relative to the baseline Parallel Embedding Fusion system which gives 0.5435 in terms of a-DCF. Green values indicate performance gains, while red values denote performance degradation.

	pooled	-	codec-1	codec-2	codec-3	codec-8	codec-9	codec-10	codec-11
pooled	1.32	-13.52	-7.49	-12.29	-4.69	2.27	14.09	20.12	-15.85
A17	-29.77	-78.21	-40.47	-65.67	-36.24	-27.56	-13.23	-5.24	-58.02
A18	15.55	24.92	13.93	11.91	13.06	11.87	25.41	27.92	12.07
A19	5.24	1.22	2.61	0.02	2.67	6.55	7.32	6.12	2.76
A20	7.34	15.69	5.66	3.49	6.60	7.62	14.23	24.84	6.38
A21	-9.50	-6.94	-12.56	-38.58	-9.72	-13.71	-4.63	1.01	-45.32
A22	3.67	15.16	-4.38	-19.32	2.46	-2.60	11.04	-22.60	-29.45
A23	21.76	30.68	18.69	10.11	19.86	7.93	12.64	22.93	-7.04
A24	12.37	-19.91	-14.35	0.86	-2.63	6.77	28.08	32.64	-19.98
A25	43.01	35.59	30.72	31.56	40.02	41.26	51.17	51.03	20.68
A26	29.15	33.86	22.75	15.18	29.38	23.47	36.86	37.83	10.36
A27	-13.21	-69.92	-36.52	-20.75	-23.16	-0.07	15.77	21.45	-24.37
A28	-0.62	-1.44	-3.43	-2.84	-6.78	-7.94	6.17	8.78	-12.60
A29	0.003	0.41	-17.04	-11.68	-9.20	-18.87	-6.85	0.80	-50.68
A30	0.77	-22.79	-9.88	-9.10	-4.22	2.69	15.37	19.18	-4.98
A31	-7.04	-53.21	-31.44	-21.38	-16.72	-6.39	7.91	23.70	-21.04
A32	-1.68	-32.87	-12.85	-15.21	-11.38	10.93	21.89	28.83	-17.01

verification and spoof detection information into the SASV task. Specifically, CM embeddings were obtained using the AASIST model [18], which was trained on the training subset of the ASVspoof 5 dataset. Meanwhile, ASV embeddings were extracted using the ECAPA-TDNN model [19], which was pre-trained on the VoxCeleb2 dataset. The ASV extractor produces 192-dimensional enrollment and test embeddings (e_{ASV}^{enr} and e_{ASV}^{tst}) while CM embeddings for enrollment and test utterances (e_{CM}^{enr} and e_{CM}^{tst}) extracted from the AASIST model have a dimensionality of 160.

To train the SASV models described in Section 2, embeddings from both extractors were fed into fully connected layers. The proposed architecture consists of three hidden layers followed by an output layer in each branch. The hidden layers contain 256, 128, and 64 nodes, respectively, while the output layer consists of a single neuron. ReLU activation was applied in all hidden layers, except for the output layer, where sigmoid activation was used for classification. The models were trained for 100 epochs using the Adam optimizer with a learning rate of 0.0001.

For performance evaluation, we adopted the architecture-agnostic detection cost function (a-DCF) [20], which serves as the primary metric for the ASVspoof 5 Challenge Track 2 condition. Consequently, a-DCF was used as the primary performance measure in our experiments.

4. Results

In the ASVspoof 5 challenge, the organizers provided two baseline systems: B03, which employs a log-likelihood ratio (LLR)-based fusion of automatic speaker verification (ASV) and countermeasure (CM) scores, and B04, which utilizes a MFA-Conformer model to extract a single embedding from the input waveform and produce a unified SASV score. The B03 and B04 systems achieved a-DCF values of 0.5741 and 0.6806, respectively [17]. In contrast, our baseline system, which adopts a parallel fusion approach (as described in Sec. 2.1), achieved a lower a-DCF value of 0.5435. This result demonstrates the effectiveness of the parallel fusion approach over both traditional score fusion (B03) and end-to-end SASV architectures (B04).

The results obtained using various parameter-sharing con-

figurations, as described in Sec. 2, are summarized in Table 1. Overall, the parameter-sharing approaches outperform the challenge baselines (B03 and B04). Among the different configurations, the best SASV performance is achieved by Setup-4, where statistical descriptors of the third hidden layer output from the parallel branch processing CM embeddings are concatenated with the SASV features learned from the ASV embeddings. This setup yields an a-DCF value of 0.5364, which is lower than both the challenge baselines and our baseline system. Additionally, applying score fusion to the various parameter-sharing configurations (as shown in the last two columns and the last row of Table 1) is generally found to further enhance SASV performance.

One could argue that the parameter-sharing techniques provide only limited improvement over the baseline system. To gain deeper insights, Table 2 presents the relative improvements or degradations for each attack type and codec in the evaluation set, using the best-performing configuration (Setup-4, $i = 3$) compared to the baseline parallel fusion approach. The results indicate that for certain attack types (e.g., A18, A19, A20, A23, A25, and A26) and codecs (8-10), parameter sharing leads to considerable improvements in SASV performance.

5. Conclusion

This study highlights the importance of parameter sharing strategies in enhancing the performance and robustness of spoof-aware speaker verification (SASV) systems. By enabling effective information exchange between automatic speaker verification (ASV) and countermeasure (CM) components, parameter sharing facilitates the development of systems that are not only more accurate but also resilient to spoofing attacks. The proposed approaches demonstrate that integrating knowledge from both ASV and CM domains through parameter sharing leads to more reliable verification systems. This work underscores the potential of multi-task learning frameworks and emphasizes that parameter sharing can serve as a fundamental building block for future advancements in secure and efficient speaker verification technologies.

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